

6(a)(i)	energy is dissipated in the internal resistance/ r	B1
6(a)(ii)	1. $I = Q/t$	C1
	$= 750 / 1500$ $= 0.50 \text{ A}$	A1
	2. $V = W/Q$ or $V = W/It$	C1
	$= 5700 / 750$ or $5700 / (0.50 \times 1500)$ $= 7.6 \text{ V}$	A1
	or	
	$V = P/I$ and $P = W/t$	(C1)
	$V = 3.8 / 0.50$ $= 7.6 \text{ V}$	(A1)
	3. $r = (7.8 - 7.6) / 0.50$	C1
	$= 0.40 \Omega$	A1
6(b)(i)	90 Ω and 45 Ω resistors shown connected in parallel	B1
6(b)(ii)	the resistors connected in parallel labelled as 90 Ω and 45 Ω with the other resistor labelled as 20 Ω	M1
	V_{OUT} or 3.6 V labelled across the 20 Ω resistor	A1